PROJECT REPORT
TRANSACTION MONITORING ENGINE

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Database: SQLite
(Simulated AML Transaction Monitoring Environment)

1 Executive Summary

This project implements rule-based AML monitoring scenarios designed to identify transaction activity requiring enhanced review within a simulated banking environment.

A standardized analytical dataset is established through date and currency standardization, together with integration of customer and jurisdiction risk indicators to support consistent monitoring execution.

Using this consolidated dataset, monitoring scenarios are executed to detect activity meeting predefined monitoring conditions, including:

- PEP and high-risk customer transaction activity
- High-value and cumulative transaction monitoring
- High-risk jurisdiction transaction exposure
- Cash aggregation activity
- Structuring behaviour detection
- Transaction velocity monitoring

All scenarios are executed through SQL-based monitoring logic and generate alert-oriented outputs aligned with operational transaction monitoring workflows. The report documents monitoring behaviour strictly based on observable system outputs without investigative interpretation beyond rule triggers.

2 Data Preparation and Monitoring Framework

2.1 Transaction Data Standardization (Currency and Date Alignment)

Transaction data is standardized to ensure consistent execution of monitoring scenarios.

- **Currency Standardization**
  All transaction amounts are converted into EUR using fixed ECB reference rates, enabling consistent threshold testing and aggregation across monitoring rules. Production environments would typically apply daily FX rate tables.
- **Date Standardization**
  Transaction dates are converted to ISO SQL date format (YYYY-MM-DD) to support chronological analysis, daily aggregation, and velocity monitoring.

These steps ensure monitoring scenarios operate on comparable transaction values and aligned timelines.

2.2 Customer Risk Profiling Integration

Customer risk attributes are integrated into the standardized transaction dataset to support risk-based analysis.

Customer-level indicators, including PEP status and derived customer risk category (High / Medium / Low), are joined to transaction records using Customer ID. Risk categories are sourced from the customer risk model developed in Project 1, ensuring consistency across projects.

This integration enriches transaction records with customer risk context within the consolidated monitoring dataset.

2.3 Consolidated Monitoring Dataset Construction

A consolidated monitoring dataset is constructed by combining standardized transaction data with integrated customer and jurisdiction risk attributes.

The dataset serves as the unified analytical base for monitoring execution, ensuring each transaction record contains normalized transaction values, aligned dates, and associated risk indicators required for rule evaluation.

This consolidated structure enables consistent rule application and supports alert-oriented outputs generated by the monitoring scenarios.

3 Monitoring Scenario Framework

3.1 Monitoring Architecture

The monitoring framework applies rule-based transaction monitoring executed through SQL. Each scenario evaluates transaction activity against predefined conditions to identify activity requiring further review, generating alert-oriented outputs when monitoring thresholds are met.

3.2 Scenario Coverage and Classification

Monitoring scenarios are grouped by objective to provide structured coverage of key financial crime risk indicators:

- High-risk customer activity
- High-value and cumulative transactions
- Jurisdiction risk exposure
- Cash aggregation activity
- Structuring behaviour
- Transaction velocity

This classification organizes related rules under common risk themes, supporting consistent interpretation of alert outputs.

3.3 Alert Generation Approach

Alerts are generated when transaction activity meets predefined scenario conditions and thresholds. Each rule trigger produces an alert identifying transactions or customer activity requiring analyst review.

Alerts represent review candidates and do not constitute investigative conclusions.

4 Monitoring Results and Observations

This section presents summarized monitoring results. Sample extracts are provided in the Appendices; full monitoring outputs are supplied separately as supporting files.

4.1 High-Risk Customer Activity Monitoring

High-Risk Customer Activity Monitoring captures transactions conducted by PEPs and High-risk customers identified under the customer risk model.

Sample output extract is provided in Appendix A (Table A.1).

Table 4.1 – High-Risk Customer Activity Monitoring Summary

Cust ID	Cust Risk Category	Cust Country Risk Level	Alert Tx Count	Total Tx Amt (EUR)	Average Tx Amt (EUR)	High-Risk Country Tx Count	Primary Alert Type
C002	Low	Low	3	12014.21	4004.74	0	Cash Activity
C003	Medium	Low	4	41998.79	10499.7	1	High-Risk Geography
C004	Low	Low	3	10184.08	3394.69	0	Mixed Activity
C008	High	Low	4	576000.57	144000.14	1	High-Risk Geography
C010	Low	Low	5	19704.12	3940.82	1	High-Risk Geography
C023	Medium	Low	5	70200.43	14040.09	2	High-Risk Geography
C026	High	Low	3	53344.18	17781.39	0	Mixed Activity
C031	High	Low	2	33109.63	16554.82	0	Mixed Activity
C032	Medium	Low	4	53982.12	13495.53	1	High-Risk Geography
C033	Medium	High	4	15880.64	3970.16	1	High-Risk Geography
C034	Medium	Medium	5	65179.92	13035.98	1	High-Risk Geography
C039	Low	Low	2	8332.36	4166.18	1	High-Risk Geography
C040	Low	Low	5	10816.34	2163.27	2	High-Risk Geography
C042	Low	Low	2	8013.25	4006.63	0	Mixed Activity
C047	Medium	High	4	15838.51	3959.63	0	Cash Activity
C048	High	High	6	95559.47	15926.58	1	High-Risk Geography

Observations:

- Alerts are generated across Low, Medium, and High customer risk categories due to inclusion of PEP activity.
- High alert volumes are observed among certain customers despite low customer-country risk levels.
- High-risk jurisdiction exposure appears through transaction countries rather than customer domicile alone.
- Cash deposits and withdrawals appear frequently within alert activity.

4.2 Transaction Value and Aggregation Monitoring

Transaction value and aggregation monitoring identifies customers exhibiting risk through high-value transactions, cumulative account activity, and aggregated cash behaviour across monitoring scenarios.

Table 4.2 – Transaction Value and Aggregation Monitoring Summary

Cust ID	Cust Risk Category	High-Value Tx Count	Largest Tx Amt (EUR)	Cum Alert Accs	Total Cum Tx Amt (EUR)	Cash Agg. Tx Count	Total Cash Agg. Amt (EUR)	Primary Alert Type
C008	High	3	230954.25	1	576000.57	0		High-Value + Cumulative Activity
C010	Low	0		0		3	15471.44	Cash Aggregation Pattern
C011	Low	0		0		3	10147.83	Cash Aggregation Pattern
C016	Low	0		0		3	16274.19	Cash Aggregation Pattern
C045	Medium	4	254942.82	2	810046.79	0		High-Value + Cumulative Activity
C047	Medium	0		0		3	11292.08	Cash Aggregation Pattern

Observations:

- Alert activity is concentrated in customers **C008** and **C045**, driven by repeated high-value transactions and significant cumulative account exposure.
- **C045** demonstrates multi-account cumulative activity, indicating concentrated fund movement consistent with layering risk indicators.
- Cash aggregation alerts (**C010, C011, C016, C047**) reflect repeated smaller cash transactions breaching monitoring thresholds.
- Results show clear separation between high-value/cumulative risks and cash aggregation behaviour, confirming complementary scenario coverage

4.3 Behavioural Pattern Monitoring

Behavioural monitoring scenarios detect structuring activity and abnormal transaction velocity using defined time-window thresholds.

Table 4.3 – Behavioural Pattern Monitoring Results

Monitoring Scenario	Detection Logic	Monitoring Window	Alerts Generated
Structuring Detection (7-Day Rolling Window)	Multiple cash deposits within monitoring window	7 days	0
Wire Transfer Velocity Detection (1-Day Rolling Window)	Multiple wire transfers within monitoring window	1 day	0

Observations:

- No customers triggered structuring or velocity alerts during the monitoring period.
- Results confirm effective scenario execution and establish a behavioural monitoring baseline under current thresholds.
- Observed risks across monitoring outcomes remain transaction-value driven rather than behaviourally driven.

4.4 Geographic Risk Exposure Monitoring

Geographic risk monitoring identifies transactions involving high-risk jurisdictions based on country risk classifications and transaction exposure levels.

Sample output extract is provided in Appendix A (Table A.4).

Table 4.4 – Geographic Risk Exposure Monitoring Summary

Cust ID	Cust Risk Category	High-Risk Country Tx Count	Total High-Risk Country Tx Amt (EUR)	Distinct High-Risk Countries	Primary High-Risk Country
C001	Low	2	24233.11	2	Russia
C003	Medium	1	12736.07	1	Nigeria
C005	Low	1	6608.16	1	Nigeria
C006	Low	1	6473.77	1	Nigeria
C007	Low	1	6554.49	1	Russia
C008	High	1	109067.52	1	Russia
C009	Low	2	7924.22	2	Nigeria
C010	Low	1	1913.81	1	Nigeria
C011	Low	1	4507.42	1	Russia
C013	Low	1	4915.24	1	Russia
C016	Low	1	3470.49	1	Russia
C019	Low	2	7341.43	1	Nigeria
C021	Low	1	5805.67	1	Russia
C023	Medium	2	29799.17	2	Nigeria
C024	Low	1	15229.92	1	Nigeria
C027	Low	1	4158.69	1	Nigeria
C028	Low	1	5868.78	1	Russia
C029	Medium	2	26570.2	2	Nigeria
C030	Low	2	12650.18	1	Russia
C032	Medium	1	16213.05	1	Russia
C033	Medium	1	6040.12	1	Nigeria
C034	Medium	1	13104.46	1	Russia
C036	Low	1	5694.02	1	Nigeria
C037	Low	1	3690.65	1	Russia
C039	Low	1	2609.37	1	Russia
C040	Low	2	4766.63	1	Nigeria
C041	Low	2	21013.37	1	Nigeria
C044	Medium	1	12801.52	1	Nigeria
C045	Medium	1	167240.68	1	Russia
C046	Low	1	5742.4	1	Nigeria
C048	High	1	18711.74	1	Nigeria
C050	Low	1	13212.95	1	Russia

Observations:

- High-risk country activity is widely distributed across customers, with most exposures occurring as isolated transactions rather than recurring patterns.
- Transactions are predominantly linked to Russia and Nigeria, indicating consistent jurisdictional risk drivers within the dataset.
- Financial exposure is concentrated in C045 and C008, where transaction values materially exceed peer activity levels.
- Most customers transact with a single high-risk jurisdiction, enabling clear identification of primary geographic risk exposure.

5 Monitoring Limitations and Data Constraints

This analysis is based on a synthetic dataset designed to simulate realistic AML risk patterns rather than represent actual customer populations. Certain risk characteristics and transaction distributions are intentionally calibrated to ensure sufficient alert generation across monitoring scenarios.

The monitoring framework applies predefined rule-based detection logic and does not incorporate adaptive behavioural modelling, peer-group analysis, or real-time monitoring capabilities. Scenario outputs rely solely on structured transaction attributes and do not incorporate external intelligence sources or investigative context, reflecting the constraints of a controlled analytical environment and areas for future enhancement.

6 Compliance and Operational Implications

6.1 Operational Strengths

- Monitoring scenarios align with rule-based transaction monitoring practices used in financial institutions.
- Combination of customer risk attributes and transactional behaviour supports risk-based alert prioritisation.
- Coverage across value, behavioural, and geographic risk indicators provides layered AML detection capability.
- Modular SQL design enhances auditability, reproducibility, and investigative usability.

6.2 Recommended Enhancements

1. Incorporate account lifecycle indicators (e.g., account age and activity maturity).
2. Enhance behavioural analytics through transaction velocity and peer comparison metrics.
3. Expand cross-border and jurisdiction exposure risk scoring.
4. Integrate sanctions and adverse-media screening controls as a dedicated monitoring layer.

7 Conclusion

This project extends the prior KYC risk profiling framework through implementation of rule-based transaction monitoring aligned with standard AML control practices. The monitoring framework provides:

- detection of high-risk customer activity, transaction value anomalies, behavioural patterns, and geographic risk exposure
- aggregated alert outputs supporting investigation prioritization
- transparent and auditable monitoring logic consistent with operational compliance environments
- a scalable foundation for future behavioural analytics and sanctions screening enhancements

The monitoring design remains rule-driven, explainable, and operationally interpretable, reflecting practical financial crime monitoring standards.

8 Appendices

The Appendices contain sample extracts of monitoring outputs referenced in Section 4. Full monitoring outputs are supplied separately as supporting files.

Dates in supporting CSV files follow ISO 8601 format (YYYY-MM-DD); display may vary depending on spreadsheet software settings.

Appendix A – Monitoring Output Extracts

Table A.1 – High-Risk Customer Activity Monitoring Output Extract

cust_id	acc_id	full_name	pep_status	cus_risk_cat	cust_country	cust_country_risk	clean_tx_date	tx_amt_eur	tx_type	is_cash	tx_country	tx_country_risk	alert_reason
C002	A052	Customer_2	Y	Low	United Kingdom	Low	2023-10-06	5461.71	Cash Deposit	1	France	Low	PEP Activity
C002	A052	Customer_2	Y	Low	United Kingdom	Low	2023-10-06	1417.25	Wire Transfer	0	Switzerland	Low	PEP Activity
C002	A052	Customer_2	Y	Low	United Kingdom	Low	2023-11-02	5135.25	Cash Deposit	1	Switzerland	Low	PEP Activity
C003	A011	Customer_3	Y	Medium	Germany	Low	2023-01-27	12366.84	Wire Transfer	0	France	Low	PEP Activity
C003	A012	Customer_3	Y	Medium	Germany	Low	2023-02-12	7973.42	Wire Transfer	0	Germany	Low	PEP Activity
C003	A011	Customer_3	Y	Medium	Germany	Low	2024-07-02	8922.46	Cash Withdrawal	1	Switzerland	Low	PEP Activity
C003	A011	Customer_3	Y	Medium	Germany	Low	2024-09-05	12736.07	Cash Withdrawal	1	Nigeria	High	PEP Activity
C004	A056	Customer_4	Y	Low	France	Low	2023-01-28	3884.1	Cash Withdrawal	1	China	Medium	PEP Activity
C004	A056	Customer_4	Y	Low	France	Low	2023-05-14	5175.64	Cash Withdrawal	1	Germany	Low	PEP Activity
C004	A056	Customer_4	Y	Low	France	Low	2024-03-16	1124.33	Wire Transfer	0	Switzerland	Low	PEP Activity
C008	A001	Customer_8	Y	High	Switzerland	Low	2024-09-10	192608.58	Cash Withdrawal	1	France	Low	PEP + High Risk

cust_id	acc_id	full_name	pep_status	cus_risk_cat	cust_country	cust_country_risk	clean_tx_date	tx_amt_eur	tx_type	is_cash	tx_country	tx_country_risk	alert_reason
C008	A001	Customer_8	Y	High	Switzerland	Low	2024-09-11	109067.52	Cash Withdrawal	1	Russia	High	PEP + High Risk
C008	A001	Customer_8	Y	High	Switzerland	Low	2025-05-01	230954.25	Wire Transfer	0	Brazil	Medium	PEP + High Risk
C008	A001	Customer_8	Y	High	Switzerland	Low	2026-01-01	43370.22	Cash Withdrawal	1	United States	Medium	PEP + High Risk
C010	A075	Customer_10	Y	Low	Singapore	Low	2022-09-30	2820.89	Cash Withdrawal	1	France	Low	PEP Activity
C010	A075	Customer_10	Y	Low	Singapore	Low	2023-01-31	7316.19	Cash Deposit	1	United States	Medium	PEP Activity
C010	A075	Customer_10	Y	Low	Singapore	Low	2023-06-15	1913.81	Cash Deposit	1	Nigeria	High	PEP Activity
C010	A075	Customer_10	Y	Low	Singapore	Low	2023-10-20	1411.78	Wire Transfer	0	France	Low	PEP Activity
C010	A075	Customer_10	Y	Low	Singapore	Low	2024-07-04	6241.44	Cash Deposit	1	United Kingdom	Low	PEP Activity
C023	A029	Customer_23	Y	Medium	France	Low	2024-08-17	10161.33	Cash Deposit	1	France	Low	PEP Activity

Table A.2 – Geographic Risk Exposure Monitoring Output Extract

cust_id	acc_id	full_name	cust_risk_category	clean_tx_date	tx_amt_eur	tx_type	is_cash	tx_country	tx_country_risk_level	alert_reason
C001	A033	Customer_1	Low	2022-10-17	8180.6	Cash Deposit	1	Nigeria	High	High-Risk Country Tx
C001	A033	Customer_1	Low	2023-07-03	16052.51	Cash Withdrawal	1	Russia	High	High-Risk Country Tx
C003	A011	Customer_3	Medium	2024-09-05	12736.07	Cash Withdrawal	1	Nigeria	High	High-Risk Country Tx
C005	A073	Customer_5	Low	2024-07-11	6608.16	Cash Withdrawal	1	Nigeria	High	High-Risk Country Tx
C006	A062	Customer_6	Low	2025-03-03	6473.77	Wire Transfer	0	Nigeria	High	High-Risk Country Tx
C007	A060	Customer_7	Low	2022-08-21	6554.49	Wire Transfer	0	Russia	High	High-Risk Country Tx
C008	A001	Customer_8	High	2024-09-11	109067.52	Cash Withdrawal	1	Russia	High	High-Risk Country Tx
C009	A069	Customer_9	Low	2025-07-12	5328.81	Cash Withdrawal	1	Nigeria	High	High-Risk Country Tx
C009	A069	Customer_9	Low	2025-11-03	2595.41	Wire Transfer	0	Russia	High	High-Risk Country Tx
C010	A075	Customer_10	Low	2023-06-15	1913.81	Cash Deposit	1	Nigeria	High	High-Risk Country Tx
C011	A070	Customer_11	Low	2023-12-14	4507.42	Cash Deposit	1	Russia	High	High-Risk Country Tx
C013	A063	Customer_13	Low	2024-07-24	4915.24	Cash Withdrawal	1	Russia	High	High-Risk Country Tx
C016	A074	Customer_16	Low	2024-04-27	3470.49	Cash Withdrawal	1	Russia	High	High-Risk Country Tx
C019	A061	Customer_19	Low	2025-05-05	2340.78	Wire Transfer	0	Nigeria	High	High-Risk Country Tx
C019	A061	Customer_19	Low	2025-09-11	5000.65	Cash Withdrawal	1	Nigeria	High	High-Risk Country Tx
C021	A065	Customer_21	Low	2023-08-13	5805.67	Cash Deposit	1	Russia	High	High-Risk Country Tx
C023	A029	Customer_23	Medium	2025-03-10	15745.15	Wire Transfer	0	Nigeria	High	High-Risk Country Tx
C023	A029	Customer_23	Medium	2026-01-01	14054.03	Cash Deposit	1	Russia	High	High-Risk Country Tx
C024	A014	Customer_24	Low	2026-01-01	15229.92	Wire Transfer	0	Nigeria	High	High-Risk Country Tx
C027	A057	Customer_27	Low	2023-11-30	4158.69	Wire Transfer	0	Nigeria	High	High-Risk Country Tx